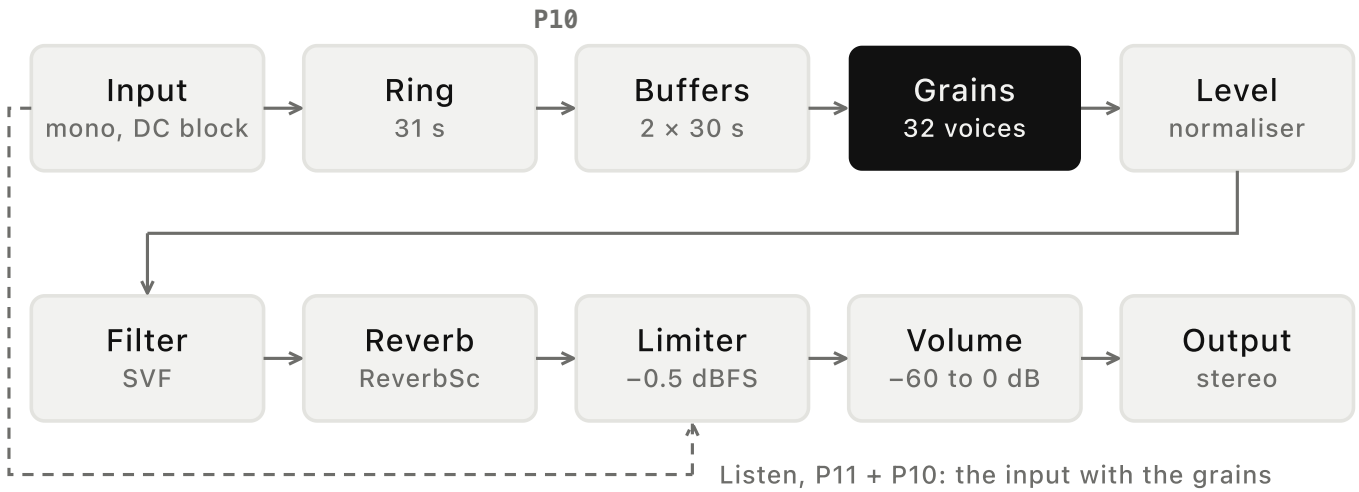
Engine

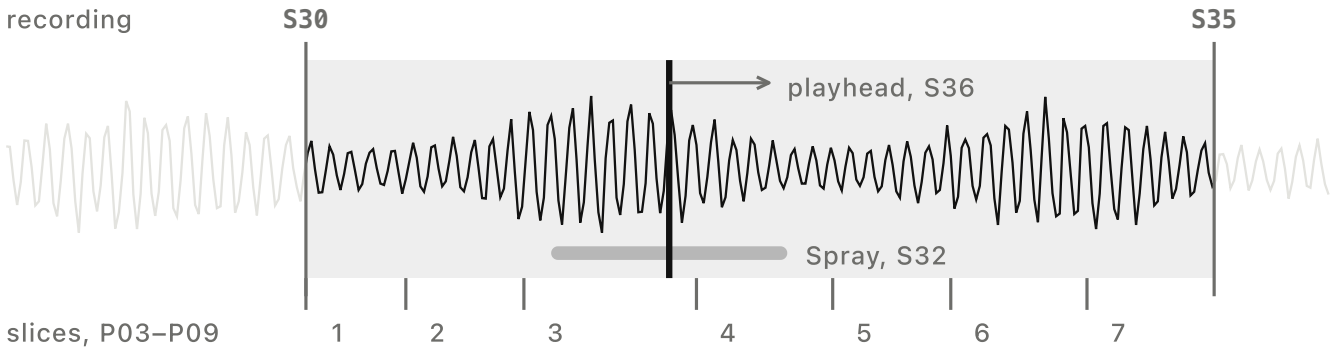
How a recording becomes grains, and how they fall.

01 SIGNAL PATH



Both jacks are summed to mono, −3 dB each. The ring takes the input all the time. P10 copies from it into the buffer not playing, and the grains read the buffer playing.

02 RECORDING



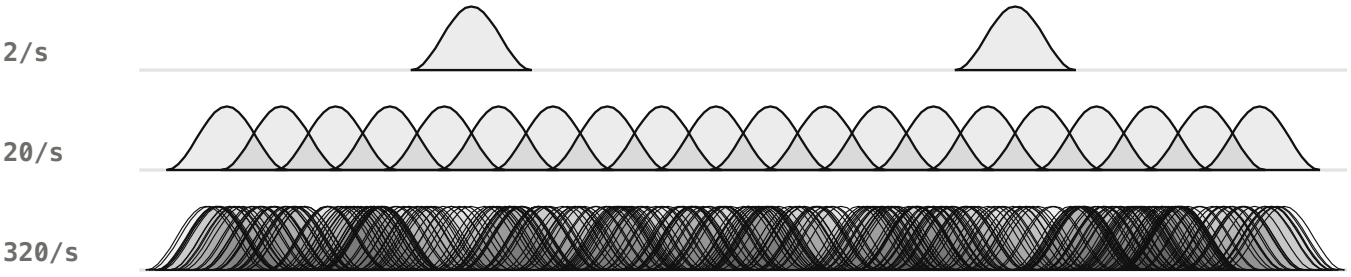
TAP	Under 0.3 s: the 8 s before it.
HOLD	From 50 ms before the touch to the release, up to 30 s.
TRIM	Silence under −60 dBFS cut from both ends.
LAND	Copied at 8×. Over 2 s new grains read it with rising odds.
SLICES	Its attacks found once, in 5 ms frames, the strongest 64 kept. A loop's seven slices: its start, then its strongest attacks. Too few, as a held note: even parts.

03 A GRAIN

POSITION	The playhead. With Spray, evenly round it, up to the loop.
START	Moved, at most 5.3 ms or a quarter of its length, to where it continues the grain before in phase: the signs of 512 samples compared, as Clouds' WSOLA player. Not with Spray up, whose grains are meant to differ.
READ	Four-point Hermite, as Beads. Rate 2^(n/12), ±24 semitones.
RANDOM PITCH	Up: free, to ±12 semitones. Down: octaves, up to 2 grains in 3.
SIZE	5 ms to 2 s, 100 ms at the centre. Random: part way to any.
WINDOW	Beads' envelope along Shape. At least 0.5 ms up, 10 ms down.
PAN	Spray spreads them, wider as more sound at once. Equal power.
DIRECTION	The playhead's. Reverse sets the share read the other way.

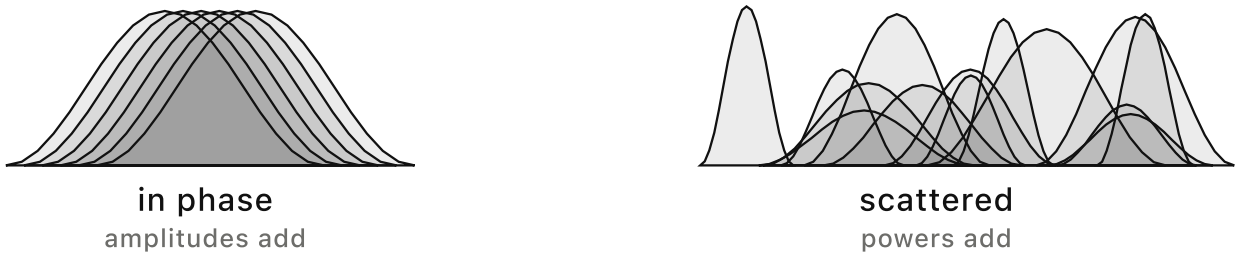


04 STARTS



Density is grains a second, 0.25 to 320, none at the bottom, whatever their size: grains sounding at once are Density × Size. Random timing takes the starts from a regular pulse to nearly Poisson, the same mean. 32 voices: when all sound, a new grain is dropped.

05 LEVEL



Grains in phase add in amplitude, scattered ones in power. A normaliser measures both every block, 5 ms smoothed, and brings the sum to the input's level with 0.9 headroom, never boosting grains that cancel. Aligned starts keep slowed grains in phase: at ½× a sawtooth's harmonics vary 0.27 dB from one 20 ms frame to the next, as at 1×. Without, 1.0 dB, and 5.6 dB in a dense random cloud.

06 TRANSPORT

SPEED	0 to 4×, the pitch kept. Frozen, a full stroke scrubs the loop.
END	S07 at the loop's end: stop, turn back, or round again.
STOP	P00 held: a tape stop over 0.6 s. Released: back to speed.
TRIGGER	P02: from the loop's start (end in reverse), a grain at once.
SLICES	P03–P09 held: each slice its own playhead, the loop waiting.

07 EFFECTS

FILTER	SVF: low-pass to 200 Hz, off at the centre, high-pass to 2 kHz.
REVERB	ReverbSc on the grains. Decay 0.5 to 16 s, up to half wet.
LIMITER	Look-ahead 5 ms, −0.5 dBFS. The grains +6 dB into it.
VOLUME	After the limiter: off, then −60 to 0 dB.

08 RUNTIME

AUDIO	48 kHz, 48-sample blocks: 1 ms. Recordings in SDRAM, mono.
CPU	15 % with no grains. 83 % at most: 32 voices, Pitch +24.
BROWSER	The same C++ core in WebAssembly, in an AudioWorklet.